

JASHANPREET SINGH

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Amritsar, Punjab, India

Education

Guru Nanak Dev University - BSc. Physics - CGPA 9.10/10 <i>Amritsar, Punjab, India</i>	2024 – Present
The Millennium School - Non Medical - 84.2% <i>Patiala, Punjab, India</i>	2022 – 2024
The Kaintal School - Secondary Education - 94.4% <i>Patiala, Punjab, India</i>	2010 – 2022

Achievements

Academic Awards

- **ICNPA** (International Conference on Nuclear Physics and Application) **Best Oral Presentation Award.** (2024)
- **JEE Mains** (Joint Entrance Examination) **AIR 231.** (2024)
- **NSEA** (National Standard Examination For Astronomy) **Scholar.** (2023)
- **IAAC** (International Astronomy and Astrophysics Competition) **Bronze Honours.** (2021)

Other Awards and Honours

- **IAAC** (International Astronomy and Astrophysics Competition) **Ambassador.** (2024 - Present)
- **ISRO** (Indian Space and Research Organisation) **Space Tutor.** (2023 - Present)
- **APY** (Astronomy Photographer of the Year) **Shortlisted.** (2021)
- **SDNP** (South Down National Park Astrophotography Competition) **Runner Up.** (2021)

Hackathons

- **Ideathon, GNDU** (Golden Jubilee Center of Entrepreneurship and Innovation) **Winner.** - 15000 INR (2025)
- **HACKOWASP7.0** (North India's Biggest Hackathon, TIET) **Runner Up.** - 150000 INR (2025)
- **Ad Astra** (National Astronomy Ideathon, NIT, Jalandhar, Punjab) **Second Runner Up.** - 5000 INR
- **Technovista 2.0** (Ideathon, Guru Nanak Dev University) **Frist Prize.** - 1100 INR (2025)
- **Zinnovatio 2.0** (Hackathon, Chandigarh University) **Top 10 of 1500 Teams.** (2025)

Publications

A Unified Deep Learning Framework for Automated Morphological Classification and Photometric Analysis using DECals Legacy Imaging (MNRAS Postitive Review)

- * Galamo is an open-source Python package for comprehensive galaxy analysis, integrating machine learning and statistical methods. It provides automated tools for morphology classification, kinematics, photometry, and spectral analysis to aid astrophysical research.
- * *Authors: Jashanpreet Singh Dingra, Hum Chand & Madhu Sudan*
- * www.galamo.org

Investigating the relationship between black hole mass and galaxy dynamics in Seyfert Type II and LINER galaxies (MNRAS Under Review)

- * Presented, for the first time, the correlation between M_{BH} and σ specifically for Seyfert Type II, an analysis made challenging due to obscuration by nuclear dust and the presence of a torus.
- * *Authors: Jashanpreet Singh Dingra and Madhu Sudan*

Recent Research Projects

Pipeline to distangle AGN flux contribution from total galaxy flux using PFVG Method (For MNRAS 2025 - Present)

- This project focusing on the study of the AGN flux variability using PFVG method from COMOS Casata Servey
- *Authors: Madhu Sudan, Jashanpreet Singh Dingra, Ankit Sarkar & Hum Chand*

Galspar & RAGA - Astronomy Pipelines for Galactic Morphological Study via GALFIT (For MNRAS 2025 - Present)

- Developed 'galspar', its primary purpose is to perform precise photometric and morphological measurements on astronomical images of galaxies stored in the standard FITS (Flexible Image Transport System) format.
- Developed 'RAGA', for automative pipeline for statistically morphology study via GALFIT
- *Authors: Jashanpreet Singh Dingra, Hum Chand and Madhu Sudan (Department of Astronomical Sciences, Central University of Himachal Pradesh)*

Measuring Stellar Velocity Dispersion of SDSS Galaxies using a MILES Template (2025 - Present) via our Fitting Pipeline: vdisp_fit

- Developed 'vdisp_fit', a Python pipeline to measure stellar velocity dispersion in galaxies using SDSS data.
- The method involves direct spectral fitting by convolving a MILES stellar template with a Gaussian broadening function and utilizes a robust χ^2 minimization to derive kinematic parameters.
- *Authors: Jashanpreet Singh Dingra and Gulab Chand Dewangan (IUCAA)*

Deep Learning Model for Alzheimer Detection and MRI Brain Segmentation (2025)

- Engineered a convolutional neural network (CNN) to classify Alzheimer's disease stages from brain MRI scans.
- The model also performs semantic segmentation to identify and delineate key brain structures affected by the disease, such as the hippocampus, to aid in early diagnosis.
- *Authors: Jashanpreet Singh Dingra and Hardeep Kaur*

Study of Stellar Dynamics of Open Star Clusters Using the Runge-Kutta Method (2025)

- Implemented an N-body simulation using the fourth-order Runge-Kutta method to model the gravitational interactions and long-term dynamical evolution of stars within open clusters.
- Analyzed stellar trajectories and cluster stability over cosmic timescales to understand dissolution processes.
- *Authors: Jashanpreet Singh Dingra and Suprit Singh (IIT Delhi)*

Crab Pulsar Spin Down Rate Over 5 Years (2015–2023) (2024)

- Analyzed archival X-ray timing data of the Crab Pulsar from 2015 to 2023 to precisely measure its spin-down rate and braking index.
- The study confirmed the expected rotational energy loss due to magnetic dipole radiation and investigated short-term timing noise and glitches.
- *Authors: Jashanpreet Singh Dingra, Pratham Jain (IIIT Raichur), Uttakarshika (IIIT Raichur) and Manam Tiwari (BNM Institute of Technology)*

Major Computational Astronomy Contributions

Astropy - Astrophysics and astronomy python package (2025 - Present)

- *Contributed in astropy.coordinates and many more*

Conferences, Schools & Talks

4th BINA Conference, Kerala, India (2025)

AdventureX, China's Biggest Hackathon, Final Round, Hangzhou, China (2025)

Conference on Advances in Cosmology, Christ University, Bangalore (2025)

Summer School on Analysis and Statistical Modelling of Space Science Data, MCNS, Karnataka (2025)

ZTF Summer School 2025, University of Minnesota, USA (2025)

13th IIST Astronomy and Astrophysics School (IAAS), Kerala (2025)

Summer School on Gravitational-Wave Astronomy, ICTS-TIFR, Bangalore (2025)

National Conference on Active Galactic Nuclie, CUHP, India (2025)

Invited Speaker @ Star Party, The Millennium School, Patiala, Punjab (2025)

International Conference on Nuclear Physics and Its Applications, New Delhi (2024)

68th Symposium on Nuclear Physics, IIT Roorkee (2024)

Speaker at Astronomy Webinar, Astronomy Club, Kosovo (2023)

Pulsar Data Analysis, HEASARC (NASA), India (2022)

Investigations

SFR Correlation with the Mass and Metallicity of Galaxies from SDSS DR7 (2023)

Conducted an analysis revealing a positive correlation between the star formation rate (SFR) and the stellar mass of galaxies, indicating that more massive galaxies tend to have higher star formation rates. Also observed a positive correlation between galaxy mass and metallicity, suggesting that massive galaxies are typically more metal-rich.

Correlation Between the Recessional Velocity and Distance of Type Ia Supernovae (2022)

Identified a strong positive correlation between the distance of a Type Ia supernova and its recessional velocity, consistent with Hubble's Law. Notably discovered a supernova with a recessional velocity of $0.8c$, located approximately 1000–1500 Mpc away.

Simulation that Predicts Mars Oppositions (2020)

Developed a Python-based simulation to predict Mars oppositions over the next 100 years, aiding in observational planning and planetary alignment studies.

24-Hour Sun Analysis from SDO Data (2020)

Analyzed solar activity using 24-hour observational data from the Solar Dynamics Observatory (SDO) on December 26, 2020, focusing on sunspot activity and solar flares.

Extra Projects

Web Development: International Music Academy - Gurmat Sangeet Taksal (2024)

Web Development: Religious Place - Gurdwara Saheedgarh Sahib, Hamilton, Ontario, Canada (2024)

Rocket Flight Computer (2023)

Capable of telemetry up to 7 km, equipped with an accelerometer, gyroscope, temperature sensors for the abort system, GPS modules, barometric sensors, servos, and biological slides for microbial collection from the toposphere.

Other International Publications

Shortlisted in International Astronomy Photographer of the Year Award (2021)

Published in : The Forbes, The BBC, The Royal Museums Greenwich.

South Down National Park International Release (2021)

Leadership / Social Work

Recognized by the Municipal Commissioner for heroically saving a child during a bus accident.

Member of the Art of Living's Punjab Flood Relief Team.

Tech Head of Guru Nanak Dev University Physics Club

Founder of Dingraastro Club, a global astronomy community with over 200 members.

Active Member of Over 7 Astronomy Clubs Worldwide

Technical Skills

Languages: Julia, Matlab, Python, R, HTML, CSS, JavaScript, PHP, C++ and SQL

Skills: Astronomy Data Analysis (Aladin, TopCat, DS9, VIREO, IRAC, etc), Scientific Educator, Web Development and Cooking